

[ENTER COMPANY NAME]

QUALITY MANAGEMENT SYSTEM

STANDARD OPERATING PROCEDURE

Change Control System

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Review Cycle	Annual (Every 12 Months)
Applicable Regulatory Standards	ISO 13485:2016 Clause 4.2.4 & 7.3.9 FDA 21 CFR Part 820.30(i) & 820.70(b) EU MDR 2017/745 Annex IX

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1.0 Purpose

This Standard Operating Procedure (SOP) establishes and defines a formal, validated system for managing and controlling changes to any aspect of product design, production documentation, manufacturing processes, utility software, labeling configuration, raw materials, or equipment validation states. The change control program prevents unintended degradation of device safety, efficacy, or regulatory certification statuses by requiring thorough cross-functional pre-impact assessment, structured re-verification protocols, and explicit pre-implementation authorization.

2.0 Scope

This procedure governs all modifications proposed subsequent to the establishment of formal baseline QMS infrastructure or after the release of design components to the Device Master Record (DMR). The boundary limits of change control encompass:

- **Design Changes:** Modifications to mechanical specifications, geometric layouts, functional materials, embedded firmware, or clinical interface requirements.
- **Process Changes:** Variations in cleanroom workflow arrangements, environment setpoints, critical chemical compound formulations, or standard manual assembly procedures.
- **Equipment and Tooling Changes:** Maintenance shifts affecting critical parameters, production tooling adjustments, or complete system infrastructure updates.
- **Documentation Changes:** Strategic revisions to inspection instructions, corporate QMS policies, supplier validation manuals, or raw material certificates of analysis boundaries.

3.0 Responsibility and Authority

3.1 Change Initiator

Any technical professional or quality technician who identifies an actionable opportunity for variation or remediation. Responsible for generating the change justification, specifying explicit technical target areas, and gathering technical references to aid evaluation panels.

3.2 Change Control Board (CCB)

A multi-disciplinary corporate committee comprised of permanent representatives from Quality Assurance, Regulatory Affairs, Manufacturing Engineering, Operations, and Product Design. The board holds absolute authority to approve, defer, or reject proposed change actions based on compliance criteria.

3.3 Regulatory Affairs Specialist

Responsible for assessing whether a change triggers mandatory external reporting to regulatory authorities, such as a major variation notification to a Notified Body (EU) or a new 510(k) file or PMA supplement submission to the FDA.

4.0 Procedure

4.1 Classification of Proposed Changes

To optimize administrative oversight, all change proposals must be categorized according to potential impact severity into three distinct classifications:

Classification	Impact Criteria	Approval and Execution Mandate
Level I (Minor)	Clerical fixes, typo corrections, editorial clarifying remarks. No impact on form, fit, or function.	Requires QA Manager signature. Does not demand full CCB panel evaluation session.
Level II (Moderate)	Minor equipment replacements, testing optimization adjustments, alternate raw material vendor additions with equivalent material properties.	Requires departmental head and QA/RA signatures. Full impact assessment sheet execution required.
Level III (Major)	Altering device geometry, physical or chemical material composition change, sterilization process change, or core firmware algorithms.	Full CCB assembly required. Mandates exhaustive protocol validation and explicit external Notified Body / FDA approval before material release.

4.2 Change Management Workflow

The structured change control lifecycle follows an unyielding sequential progression designed to catch risks prior to physical operational instantiation:

Phase 1: Initiation and Request Formulation

The Change Initiator populates the Change Request Form (CRF). This document must articulate the clear technical reason for change, current state vs. proposed state details, and an initial technical analysis.

Phase 2: Comprehensive Impact Assessment

The request is routed to the CCB panel. Review fields include evaluation of biological compatibility effects, process optimization stability, shelf-life integrity risk, and impact on historical risk analysis records (e.g., FMEA update requirements).

Phase 3: Re-Verification and Validation Execution

Major and moderate changes must have associated validation or testing verification protocols drafted and approved prior to physical modification. Testing results must be summarized in a formal technical report for board evaluation.

REGULATORY WARNING: Under no circumstances shall product manufactured under temporary verification testing states be released into commercial inventory until the final implementation phase of the Change Request is fully closed and signed off.

4.3 Regulatory Assessment Matrix

The Regulatory Affairs Specialist must formally document a regulatory assessment for every Level II and Level III change. This assessment determines whether the modification significantly affects the safety or effectiveness of the device, or alters the intended use statement. If an external submission is triggered, the change remain in a "Pending Regulatory Clearance" holding state, and no affected commercial product can be introduced into interstate or global commerce until official clearance is received.

4.4 Implementation and Closure

Once validation tasks are verified as passing, documentation is updated, training is deployed, and regulatory clearances are obtained, the QA Manager can execute final closure. The change record status is updated to "Closed/Effective" in the Master Change Control Index.

5.0 Documentation and Auditing Metrics

All finalized Change Request files, protocols, raw performance data, and regulatory assessments are quality records and must be linked to the master Device Master Record (DMR). These files must be retained securely for the active lifetime of the device family plus an additional five (5) years, making them readily retrievable during regulatory audits or corporate inspections.

6.0 Associated Reference Materials

- **SOP-QA-001:** Document and Data Control System
- **SOP-QA-007:** Risk Management Program (ISO 14971 Integration)
- **Form-QA-002A:** Change Request Form (CRF)
- **Form-QA-002B:** Change Control Master Tracking Index

7.0 Revision History Log

Revision Level	Effective Date	Author / Originator	Detailed Description of Modifications
00	[Date]	[Name], QA	Initial baseline release of formalized Change Control System Standard Operating Procedure for global QMS compliance.